



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Subspaces, Separable Spaces

Dense Set and Perfect Set

FEBRUARY 08, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

For updates subscribe on  [@ParthoSutraDhor](#)

Subspaces

Subspace and Induced Metric

Let (X, d) be a metric space and let Y be a non-empty subset of X . The restriction of the metric d to $Y \times Y$, denoted by d_Y , is a metric on Y called the *induced metric*. The metric space (Y, d_Y) is called a *subspace* of (X, d) .

Examples of Subspaces

Verify that:

1. the closed unit interval $[0, 1]$ and the set of all rational numbers $\mathbb{Q}$ are subspaces of $\mathbb{R}$ with the usual metric;
2. the unit circle, the closed unit disc, and the open unit disc are subspaces of the metric space $(\mathbb{C}, d)$ of complex numbers;
3. the real line $\mathbb{R}$ itself is a subspace of the complex plane $\mathbb{C}$.

Open Sphere in a Subspace

Let (X, d) be a metric space, let $Y \subseteq X$, and let $y \in Y$. The open sphere in the subspace (Y, d_Y) centered at y with radius $r > 0$ is denoted by $S_r^Y(y)$ and is defined by

$$S_r^Y(y) = \{x \in Y : d_Y(x, y) < r\}.$$

Relation Between Open Spheres

Show that for every $y \in Y$ and every $r > 0$,

$$S_r^Y(y) = S_r(y) \cap Y,$$

where $S_r(y)$ denotes the open sphere in (X, d) .

💡 Open Sets in Subspaces

Let (X, d) be a metric space and let $Y \subseteq X$. A subset A of Y is open in (Y, d_Y) if and only if there exists a set G open in (X, d) such that

$$A = G \cap Y.$$

🔗 Open Sets in a Subspace vs Ambient Space

Show that if $Y \subseteq X$ and a subset of Y is open in (X, d) , then it is also open in the subspace (Y, d_Y) . Show by examples that the converse need not be true.

❓ Open in a Subspace may Not be open in Superspace

Let $X = \mathbb{R}$ with the usual metric and let

$$Y = [0, 1].$$

Then the open sphere in (Y, d_Y) centered at 0 with radius $\frac{1}{2}$ is

$$S_{1/2}^Y(0) = [0, \frac{1}{2}),$$

which is open in Y but not open in $\mathbb{R}$. Note that Y itself is not open in $\mathbb{R}$.

❓ Example: Real Line vs Complex Plane

Show that an open interval of the real line is not an open subset of the complex plane $\mathbb{C}$ with the usual metric.

Dense Sets

Dense Set

A subset A of a metric space (X, d) is said to be *dense* (or *everywhere dense*) in X if the closure of A is X , that is,

$$\overline{A} = X.$$

Nowhere Dense Sets

Nowhere Dense Set

Let A be a subset of a metric space (X, d) . The set A is said to be *nowhere dense* in X if the interior of its closure is empty, that is,

$$\text{int}(\overline{A}) = \emptyset.$$

Basic Properties of Nowhere Dense Sets

Show that:

1. every finite subset of X is nowhere dense;
2. every subset of a nowhere dense set is nowhere dense;
3. A is nowhere dense in X if and only if $\overline{A}$ is nowhere dense in X .

Somewhere Dense Sets and Dense-in-itself Sets

vspace10pt

Somewhere Dense Set

A subset A of X is said to be *somewhere dense* in X if it is not nowhere dense. Equivalently, A is somewhere dense in X if and only if $\overline{A}$ contains a non-empty open sphere.

Dense-in-itself Set

A subset A of a metric space (X, d) is said to be *dense-in-itself* if every point of A is a limit point of A , that is,

$$A \subseteq A'.$$

Perfect Sets

vspace10pt

Perfect Set

A subset A of a metric space (X, d) is said to be *perfect* if it is closed and dense-in-itself, that is,

$$A = A'.$$

vspace10pt

Examples of Perfect Sets

Show that every closed interval, the empty set $\emptyset$, and the whole space X are perfect sets.

② Cantor Set is Perfect

Show that the Cantor set is a perfect set.



Separable Metric Space

Separable Metric Space

A metric space (X, d) is said to be *separable* if there exists a countable subset of X which is dense in X .

② Euclidean Space is Separable

Show that the Euclidean space $\mathbb{R}^n$ is separable.



❓ ℓ^p is Separable

Show that the metric space ℓ^p of all sequences (x_n) of real numbers such that $\sum_{n=1}^{\infty} |x_n|^p < \infty$ with the metric

$$d_p((x_n), (y_n)) = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{1/p}$$

is separable for every $1 \leq p < \infty$.

② ℓ^∞ is Not Separable

Prove that the metric space ℓ^∞ of all bounded sequences with the supremum metric is not separable.

Thank You!

We'd love your questions and feedback.

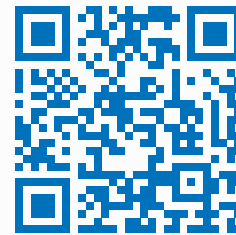
Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

 **@ParthoSutraDhor**

(Lectures, walkthroughs, and course updates)



Scan for the channel

References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.